DM IMCQ 5

Physiology, Histology, Biochemistry

KEY file

DM IMCQ 5

Histology 2 Questions

An investigator is studying tissue samples of biopsy specimens obtained during an endoscopic study from different sites along the gut, in a patient suspected of having Crohn's disease. The endoscopic study shows mucosal ulceration with the typical ‘cobblestone’ lesions, in a region of the gastrointestinal tract that is normally characterized by taeniae coli on gross inspection. Which of the following microanatomical features best characterizes this area of the gastrointestinal tract?

A. Rugae
B. Peyer’s patches
C. Paneth cells
D. Abundant goblet cells
E. Brunner’s gland
F. Microfold cells
G. Cardiac glands

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Describe the microscopic structure of the esophagus.SOM.MK.I.BPM2.3.DM.1.HCB.1206
Describe the microscopic structure of the three anatomical regions of the small intestine.SOM.MK.I.BPM2.3.DM.1.HCB.1228
Describe the microscopic structure of the different anatomical regions of the large intestine.SOM.MK.I.BPM2.3.DM.1.HCB.1246

A 32-year-old man comes to the physician because of a 6-month history of intermittent abdominal cramps and bloating. The symptoms typically occur following meals that are rich in carbohydrates. Physical examination shows no abnormalities. Genetic testing shows a mutation in a gene for an enzyme that breaks down oligosaccharides to monosaccharides for absorption. Which of the following best characterizes the localization of this enzyme?

A. Cytoplasm of hepatocytes
B. The lumen of the stomach
C. Cytoplasm of enterocytes
D. Digestive organelles of hepatocytes
E. Glands of the intestinal submucosa
F. Digestive organelles of enterocytes
G. Microvilli of the enterocytes

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Give the function(s) of the mucosa of the small intestine.SOM.MK.I.BPM2.3.DM.1.HCB.1232

DM IMCQ 5

Physiology 3 Questions

A 36-year-old man comes to the physician because of a 6-week history of intermittent watery diarrhea, abdominal cramps, and a 4-kg (8.8-lb) weight loss. His uncle had similar symptoms and died at the age of 40 years. His pulse is 88/min and blood pressure is 110/64 mm Hg. Physical examination shows dry tongue and mucus membranes. Laboratory studies show an elevated serum concentration of vasoactive intestinal peptide (VIP) and achlorhydria. Stool studies show secretory diarrhea and the absence of infection. MRI of the abdomen shows a tumor mass in the tail of the pancreas. He is diagnosed with VIPoma, a tumor that secretes VIP. Which of the following mechanisms best explains achlorhydria in this patient?

A. Activation of adenylyl cyclase in the parietal cells
B. Inhibition of epithelial chloride ion channels
C. Increased intraepithelial cAMP concentration
D. Inhibition of gastric acid secretion
E. Stimulation of pancreatic bicarbonate secretion
F. Reducing gastric somatostatin secretion

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SOM.MK.II.BPM2.3.DM.1.PHYS.1099. List the major excitatory and inhibitory motor neurotransmitters and digestive hormones in the GI tract and their key regulatory functions.

Investigators conduct a study examining the feedback and feedforward reflexes in the gastrointestinal tract. For this study, 10 healthy volunteers between 20–25-year age are selected. A balloon is inserted into the stomach of these participants, and the stomach is distended by inflating the balloon for 5 minutes. Peristaltic movements in stomach, intestines and rectum are recorded. The experiment is repeated three times with a 10-minute break between the trials. It is observed that the stomach distension produces an increased peristalsis in the distal small intestine and relaxes the sphincter between the small and large intestines. Which of the following reflexes best explains this observation seen in the participants?

A. Enterogastric reflex
B. Gastroileal reflex
C. Gastrocolic reflex
D. Ileal brake
E. Rectocolic reflex

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SOM.MK.II.BPM2.3.DM.1.PHYS.1112. Describe how distension of organs affects GI reflexes and alters responses to other regulatory inputs.

A 10-year-old girl is brought to the emergency department by her mother because of a 5-day history of fever, abdominal pain, and watery diarrhea. Three weeks ago, she had middle ear infection and was treated with antibiotics for 10 days. The mother says that she passes frequent watery stools associated with abdominal cramps. Her temperature is 38°C (100°F), pulse is 132/min, and blood pressure is 98/56 mm Hg. Abdominal examination shows distension, diffuse tenderness, and increased bowel sounds. Laboratory studies show increased bicarbonate content of the stools and presence of *Clostridioides difficle* infection. She is diagnosed with antibiotic-induced bacterial diarrhea. Which of the following additional laboratory findings is most likely in this patient?

A. Increased stool cAMP
B. Decreased stool pH
C. Increased serum sodium
D. Metabolic acidosis
E. Metabolic alkalosis
F. Increased stool bile salt

A 10-year-old girl is brought to the emergency department by her mother because of a 5-day history of fever, abdominal pain, and watery diarrhea. Three weeks ago, she had middle ear infection and was treated with antibiotics for 10 days. The mother says that she passes frequent watery stools associated with abdominal cramps. Her temperature is 38°C (100°F), pulse is 132/min, and blood pressure is 98/56 mm Hg. Abdominal examination shows distension, diffuse tenderness, and increased bowel sounds. Laboratory studies show increased bicarbonate content of the stools and presence of *Clostridioides difficle* infection. She is diagnosed with antibiotic-induced bacterial diarrhea. Which of the following additional laboratory findings is most likely in this patient?

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SOM.MK.II.BPM2.3.DM.1.PHYS.1137. Describe the related roles of fluid malabsorption in the small intestine versus colon on the potential to cause diarrheal disease.

DM IMCQ 5

Biochemistry 11 Questions

A 42-year-old man is admitted to the intensive care unit after sustaining extensive third-degree burns over 40% of his body surface area. Three days later, he is febrile and tachycardic. Laboratory studies are shown. Which of the following best explains the laboratory findings in this patient?

Laboratory Test	Patient Value	Normal Range
Plasma Glucose	235 mg/dL	70–100 mg/dL (fasting)
Serum Free Fatty Acids	1.1 mmol/L	0.2–0.6 mmol/L
Urinary Urea Nitrogen	34 g/day	12–20 g/day
Serum Ketone Bodies (β-hydroxybutyrate)	0.3 mmol/L	<0.6 mmol/L (fed state)

A. Decreased activation of hormone-sensitive lipase
B. Increased hepatic malonyl-CoA concentrations
C. Increased circulating insulin levels
D. Increased protein synthesis in the skeletal muscles
E. Decreased cortisol secretion

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SOM. MK.III.BPM2.3.DM.3.BCHM.1355 Correlate hormonal and metabolic changes (carbohydrate, protein and lipid) following trauma in the three phases: Ebb phase (Before resuscitation), Flow phase (Adrenergic-corticoid phase), Recovery phase (convalescent/ anabolic phase)

A 22-year-old man is brought to the emergency department by his friend because of a 1-day history of tiredness, vomiting and headaches. He says that this is his third time coming to the hospital and was discharged from the emergency department twice before. Physical examination shows a fruity smell of his breath. Laboratory studies show blood glucose is 400 mg/dL (70-100 mg/dL). Urinalysis shows glucose (3+) is present and ketones are present, and his urine pH is decreased. Which of the following enzymes is most likely phosphorylated in his adipose tissue at this time in this patient?

A. Hormone sensitive lipase
B. Hexokinase
C. Phosphofructokinase-2
D. Glycogen phosphorylase
E. Lipoprotein lipase

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SOM. MK.III.BPM2.3.DM.3.BCHM.1320; Describe roles of insulin and glucagon and insulin:glucagon ratio in regulation of carbohydrate, lipid and protein metabolism.

A 35-year-old woman comes to the physician because of a 6-month history of passing pale colored stools. Physical examination shows scleral icterus. Laboratory studies show highly elevated ALP and GGT levels. Which of the following additional findings is most likely present in this patient?

A. Increased serum methylmalonate levels
B. Macrocytic anemia
C. Decreased H-bonds in collagen
D. Increased prothrombin time
E. Microcytic anemia
F. Decreased activity of 1-hydroxylase
G. Decreased serum ferritin levels

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SOM. MK.III.BPM2.3.DM.3.BCHM.1302 Predict the reason for occurrence of fat-soluble vitamin deficiency in patients with fat malabsorption (cystic fibrosis, obstructive jaundice)

A 23-year-old man comes to the physician because of a 2-day history of yellowish coloration of his eyes. He has been fasting for 1-week as he wants to reduce weight. Physical examination shows jaundice. Laboratory studies are shown. Which of the following is the most likely diagnosis?

	Patient values	Reference ranges
ALT	25	10-40 U/ L
AST	32	12-38 U/ L
ALP	65	25-100 IU/ L
GGT	65	50-70 U/L
Total bilirubin	3	0.1-1.0 mg/ dL
Direct bilirubin	2.8	0.1-0.3 mg/dL

A. Gilbert syndrome
B. Dubin-Johnson syndrome
C. Sickle cell crisis
D. Gallstone impacted in ampulla of Vater
E. Bone disease
F. Chronic alcoholic cirrhosis
G. Acute viral hepatitis

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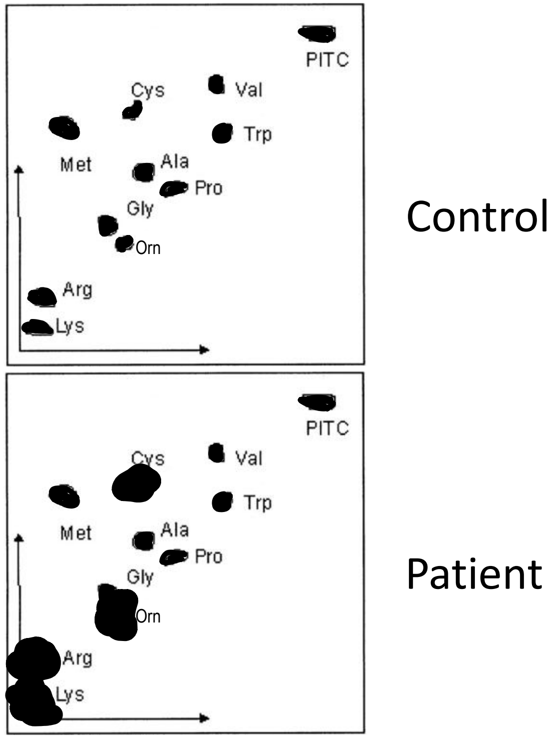
D. Gallstone impacted in ampulla of Vater
E. Bone disease
F. Chronic alcoholic cirrhosis
G. Acute viral hepatitis

SOM. MK.I.BPM2.3.DM.4.BCHM.1287; Differentiate between inherited disorders: Gilbert syndrome, Dubin Johnson syndrome and Crigler-Najjar syndromes I and II based on age of presentation, pathogenetic mechanisms and metabolic alterations

A 12-year-old girl is brought to the physician by her parents because of a 5-day history of blood in the urine and right sided flank pain. A urine sample is obtained for testing. The local hospital does not have a GC/MS spectrometer but does perform urinalysis by 2D-thin layer chromatography. The results of the test are shown. Which of the following is the most likely diagnosis?

A. Hartnup disease
B. Cystinuria
C. Homocystinuria
D. Maple syrup urine disease
E. Alkaptonuria
F. Phenylketonuria



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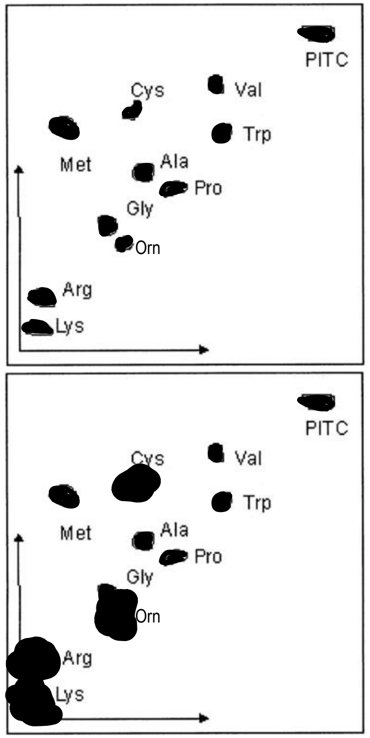

Control

Patient

SOM. MK.I.BPM2.3.DM.2.BCHM.1219; Analyze and correlate clinical features and laboratory findings to the metabolic basis of cystinuria and Hartnup's disease

A 67-year-old man comes to the physician because of a 2-day history of painful blisters on his arms and hands. He also says his urine has a reddish color. Physical examination shows bilateral, tender, dry ulceration and blisters on the dorsum of his hands. Blood and urine samples are obtained for laboratory studies. Which of the following laboratory findings is most likely in this patient?

A. Presence of bilirubin in urine
B. Elevated blood uroporphyrin-III levels
C. Yellow urine which darkens on exposure to light
D. Presence of glycosaminoglycans in urine
E. Elevated serum unconjugated bilirubin
F. Presence of porphobilinogen in urine
G. Elevated uroporphyrin-I and coproporphyrin-I in urine

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SOM. MK.I.BPM2.3.DM.2.BCHM.1271; Classify porphyrias based on clinical manifestations such as photosensitivity, abdominal pain and neuropsychiatric manifestations.
SOM. MK.I.BPM2.3.DM.2.BCHM.1276; Describe the metabolic basis for clinical features of porphyria cutanea tarda. Identify the deficient enzyme and products accumulating in blood and urine.

A. Presence of bilirubin in urine
B. Elevated blood uroporphyrin-III levels
C. Yellow urine which darkens on exposure to light
D. Presence of glycosaminoglycans in urine
E. Elevated serum unconjugated bilirubin
F. Presence of porphobilinogen in urine
G. Elevated uroporphyrin-I and coproporphyrin-I in urine

A 32-year-old man comes to the physician because of a 4-week history of watery diarrhea and bloating. Physical examination shows hyperactive bowel sounds and mild, diffuse abdominal tenderness without guarding or rebound. A jejunal biopsy specimen shows destruction of the brush border of intestinal mucosal cells. Which of the following enzymes is most likely present at this location?

A. Alpha-Amylase
B. Lipase
C. Lactase
D. Trypsin
E. Phospholipase A2

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SOM. MK.I.BPM2.3.DM.3.BCHM.1240 Outline digestion and absorption of carbohydrates. Describe reactions of disaccharidases in brush border of intestinal cells.

A 65-year-old woman comes to the physician because of a 3-month history of progressively worsening numbness and tingling in her arms and legs. Complete blood count shows MCV = 105fL (80-100). Laboratory studies show elevated serum methylmalonic acid and homocysteine levels. Which of the following is the most likely cause for this clinical presentation?

A. Dietary folate deficiency
B. Decreased dietary iron content
C. Pyridoxine deficiency
D. Decreased secretion of intrinsic factor
E. Decreased dietary copper content
F. Lead poisoning

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SOM. MK.I.BPM2.3.DM.2.BCHM.1266 Indicate role of intrinsic factor in vitamin B12 absorption

A 65-year-old woman comes to the physician for a follow-up examination. She has a 10-year history of type 2 diabetes mellitus. Her BMI is 30 kg/m^2. Laboratory studies show Hemoglobin A_{1c} is 8% (reference range:<6%) and fasting serum glucose concentration is 180 mg/dL (reference range: 70-100 mg/dL). Urinalysis shows microalbuminuria. Which of the following best explains the urinary findings in this patient?

A. Advanced glycation end products of protein
B. Decreased activity of lipoprotein lipase
C. Increased activity of hormone sensitive lipase
D. Intracellular dehydration
E. Increased LDL and low HDL levels

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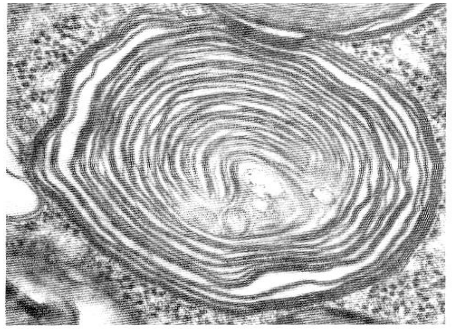
MK.I.BPM2.3.DM.3.BCHM.1369 Explain biochemical basis for mechanism of microvascular changes (sorbitol, AGE formation, role of DAG, hexosamine pathway) for nephropathy, neuropathy and retinopathy, Peripheral arterial disease and diabetic foot changes

A 2-year-old boy is brought to the physician by his mother because of a 3-month history of tripping over objects and developmental delay. Physical examination shows milestone developmental delay. Skin punch biopsy show abnormal lysosomes. Which of the following compounds is most likely accumulating in his lysosomes?

A. Sphingomyelin
B. Globoside
C. Ganglioside
D. Glycosaminoglycan
E. Glycogen
F. N-linked glycoprotein
G. O-linked glycoprotein



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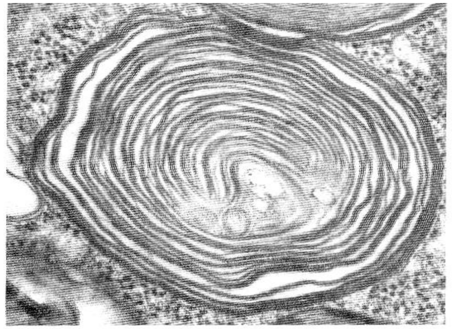
BCHM.MB.1804 Distinguish between the lysosomal storage diseases based on deficient enzyme, accumulating product, clinical features, general cell appearance and unique features

A. Sphingomyelin
B. Globoside
C. Ganglioside
D. Glycosaminoglycan
E. Glycogen
F. N-linked glycoprotein
G. O-linked glycoprotein



A 10-year-old boy is brought to the physician by his parents because of a 2-week history of severe diarrhea and rashes on his hands and feet. He is below the 60^{th} percentile for height and weight of age and sex-matched controls. Urinalysis shows elevated urinary neutral amino acids. Which of the following is the most likely diagnosis?

A. Cystinuria
B. Hartnup disease
C. Phenylketonuria
D. Maple syrup urine disease
E. Homocystinuria
F. Refsum disease

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SOM. MK.I.BPM2.3.DM.2.BCHM.1219 Analyze and correlate clinical features and laboratory findings to the metabolic basis of cystinuria and Hartnup’s disease